Fish Length Estimation with Vision Foundation Models:
Can They Beat the AutoFish CNN Baseline?

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Motivation & Question

- Fish length is a key measurement for fisheries science and catch monitoring. Manual measuring is slow and invasive.
- The AutoFish benchmark (Bengtson et al., WACVW 2025) provides images, annotations, and a CNN length-regression baseline.
- Vision foundation models promise strong general features, but do they help **precise metric regression**?

Question: Do foundation-model encoders (DINOv2, CLIP) or newer supervised encoders (ConvNeXt) improve fish length estimation over the reproduced AutoFish MobileNetV2 baseline, using identical data, splits, and metrics?

Data & Setup

- **AutoFish dataset:** 1,500 top-view images, 454 unique fish, 18,157 fish annotations, organized in 25 fish groups.
- **Official group-level split:** 15 train / 5 validation / 5 test groups. No fish appears in two splits (one duplicate fish identity was found and removed).
- **Two test regimes:** non-occluded (Set1+Set2, fish laid separately) vs. occluded (“All”, fish overlapping).
- **Metric:** mean absolute error (MAE) of predicted length in cm, lower is better. RMSE, MAPE, and R^2 are tracked as secondary metrics.

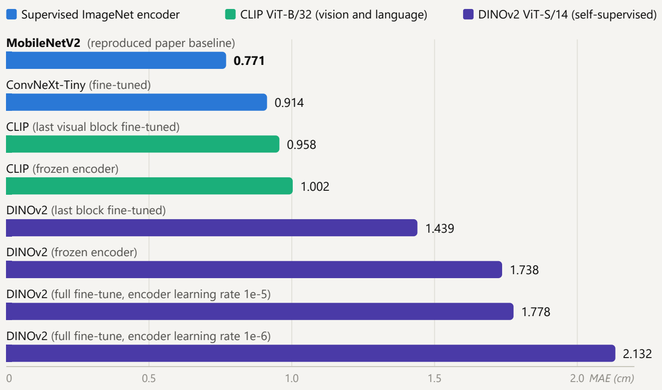
Method / Workflow

Fish crop (bounding box) → Resize 224 × 224 → Image encoder → + box size (4 values) → Regression head [512, 128, 1] → **Length (cm)**

- **Encoder comparison.** Split, input crops, target, bbox input, L1 loss, and metrics are controlled; the regression-head concept is shared, with hidden sizes adapted to each encoder.
- **Learning rate follows each encoder type:** the MobileNetV2 baseline reproduces the paper setup (Adam, 1e-3); foundation models are fine-tuned at lower rates (1e-4, and 1e-5/1e-6 for DINOv2 full fine-tune), as pretrained encoders require.
- **Encoders:** MobileNetV2 (paper baseline), ConvNeXt-Tiny, CLIP ViT-B/32 (frozen & last-block fine-tuned), DINOv2 ViT-S/14 (CLS-token and patch-token variants).

Main result: the lightweight MobileNetV2 baseline still wins

Mean absolute error (cm) on the full test set (3,759 fish, occluded + non-occluded). Lower is better.



Takeaway: every foundation-model variant trails the fine-tuned MobileNetV2 baseline. Task-adapted supervised encoders (blue) fill the top of the ranking; general-purpose features transfer only partially (CLIP) or poorly (DINOv2).

Figure: full-test MAE per model; identical split / input / head / metrics, learning rate set per encoder type (see Method).

Baseline reproduction

Non-occluded test MAE (Set1+Set2), our run vs. the AutoFish paper:

Paper (Bengtson et al.)	Our reproduction
0.62 cm	**0.633** cm

A difference of only 0.013 cm, so the baseline and our whole comparison pipeline can be trusted.

Key findings

- **Baseline reproduced** within 0.013 cm of the paper; **MobileNetV2 stays best** (0.771 cm).
- **No foundation model beats the CNN** at length; CLIP (0.958 cm) transfers far better than DINOv2.
- **Patch tokens reliably beat the CLS token** for DINOv2 (0.58 cm, 3 seeds) — best DINOv2 setup.
- **Occlusion hurts every model** (baseline 0.633 → 0.909 cm).

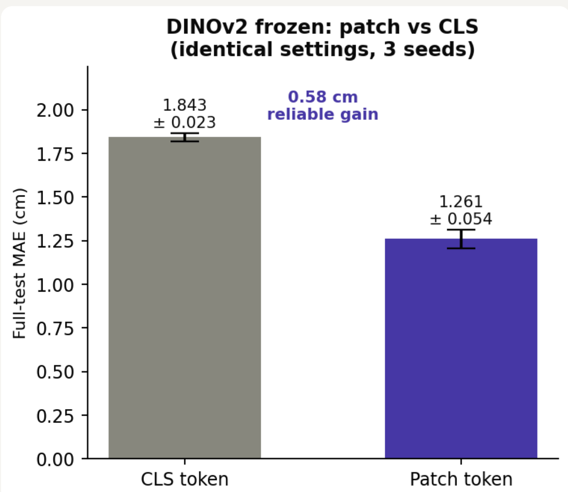
Limitations

- Single-run for the top length models (multi-seed done for the DINOv2 patch-vs-CLS comparison).
- Foundation models tested at small scale (ViT-S/14, ViT-B/32); larger variants may differ.

Next steps

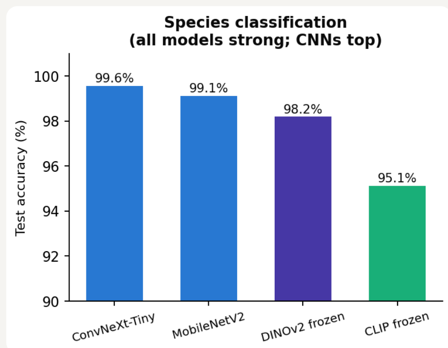
- Multi-seed the top length models; try EfficientNet-B0 and patch pooling for CLIP.
- Mask segmentation (Mask2Former-style) as a third task; DINOv3 pending weight access.

New result: patch tokens beat the CLS token



- **Controlled test, 3 seeds, identical settings.** Only the DINOv2 feature type differs.
- Patch-token pooling: **1.261 ± 0.054 cm** vs CLS token 1.843 ± 0.023 cm.
- **0.58 cm reliable gain**, non-overlapping ranges. Best DINOv2 setup; last-block fine-tune did not help.

Second task: species classification



- All encoders 95–99.6% accuracy.
- DINOv2 rises from last (length) to near-top (species): general features suit *what*, not *how big*.

Reference: S. H. Bengtson et al., “AutoFish: Dataset and Benchmark for Fine-grained Analysis of Fish,” WACV Workshops 2025 (arXiv:2501.03767). Dataset: Hugging Face *vapaau/autofish*.

Reproducibility: fixed seed (42), official group split, versioned configs per experiment, requirements.txt, saved checkpoints & per-fish prediction files. Trained on one NVIDIA RTX 5000 Ada (32 GB).

AI statement: generative AI tools were used to assist with code scaffolding and text drafting; all experiments, results, and conclusions are the authors' own work and were verified by the authors.